

BSYS 2000 Project 1:
Retail Store Data Model and ERD Explanation

Key Assumption:

1. Customers records are created only when a purchase is made.
2. Categories are created only when products exist.

Design Goal:

As an IT consultant, I designed this data model to modernize the store's operations and support structured tracking and reporting of sales, inventory, and customer data.

Entity Choices:

The core entities include Customers, Products, Categories, and Sales. I also created a Sales_items entity to represent the many-to-many relationship between sales and products.

Attribute Choices:

In addition to basic customer information, I included date of birth and join date to support targeted marketing strategies, such as birthday promotions and engagement campaigns for new customers. These new attributes might support decisions regarding which customers to target.

In the Sales entity, customer_id is defined as a foreign key, and the relationship between Customers and Sales is designed as one-to-many to represent that one customer can have multiple sales records.

In the Products entity, basic product information is included, and category_id is defined as a foreign key to associate with a specific category. The relationship between Categories and Products is one-to-many, which means one category can contain multiple products.

The Sales_items entity collects sales detailed information, including products' ID, quantity sold, and unit price. This design helps the business to identify key performance metrics such as top-selling and slow-moving products, analyze sales by category, and track month-over-month sales trends to optimize inventory levels.

Overall, the data model is designed to support informed inventory decisions and targeted marketing strategies, while providing the necessary data structure for generating monthly reports.